

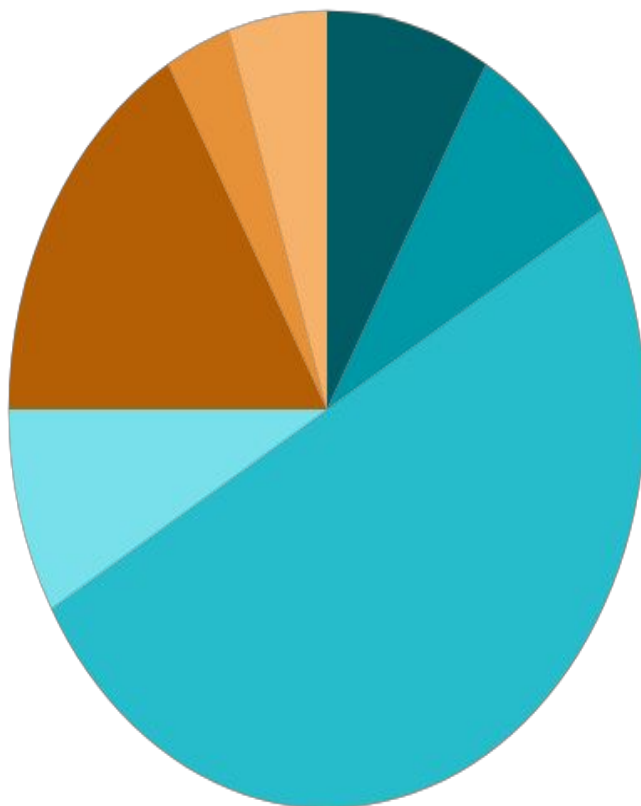


Introduction of Big Data

Outline

- What is Big Data?
- Why Big data?
- Industry Examples.
- Web Analytics
- Big Data and Marketing

Lecture Structure



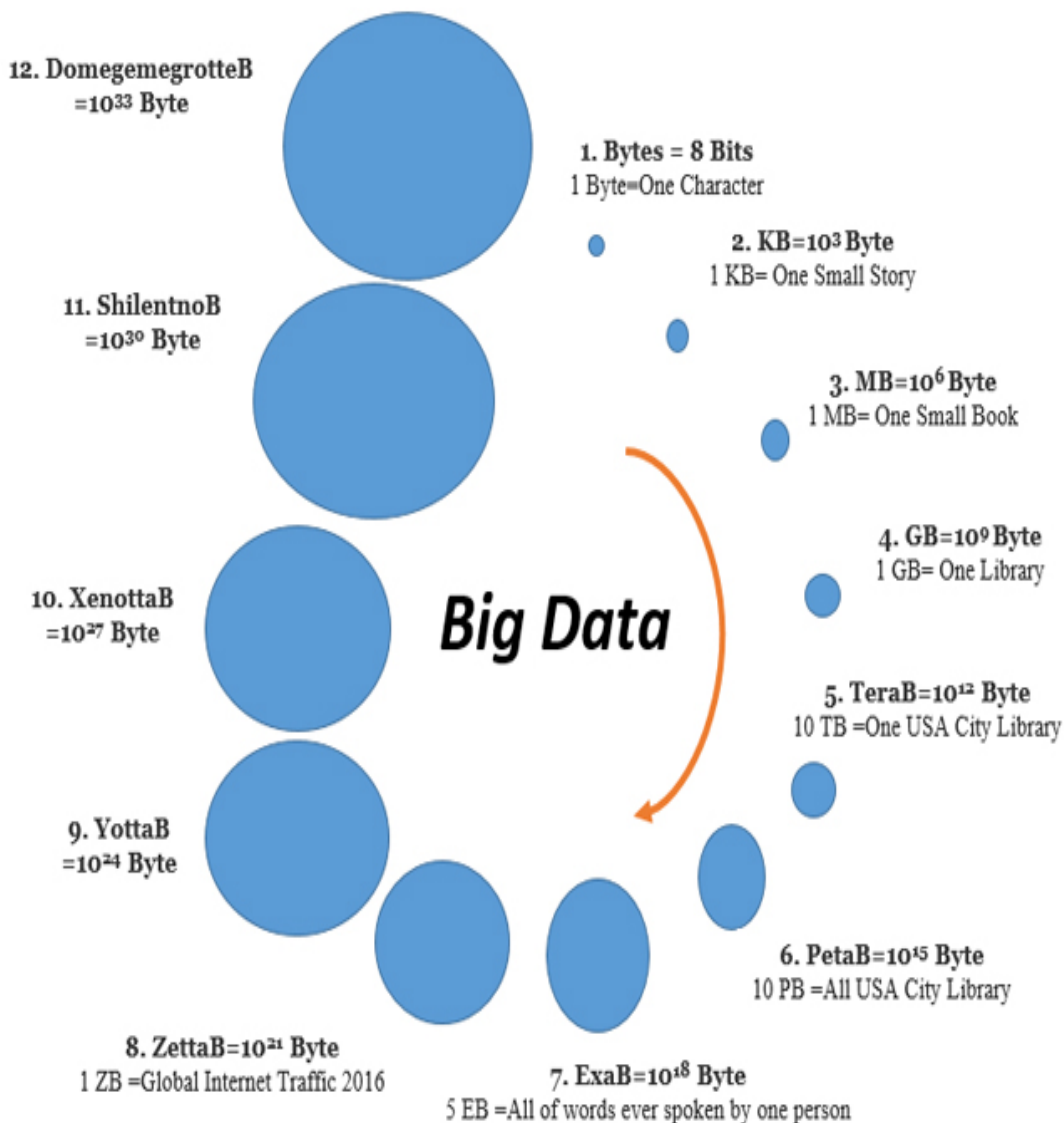
	05 Minutes	Outline & Objectives
	05 Minutes	Recap of Previous Lecture
	30 Minutes	Delivery of Actual Content as per Lesson Plan
	05 Minutes	Industry Relevance
	10 Minutes	Interactive Activities
	02 Minutes	Importance of Discipline
	03 Minutes	Recording of Attendance

What is Big Data?

- Big Data is a **massive amount of data sets** that cannot be stored, processed, or analyzed using traditional tools.
- “Big data is **high-volume**, **high-velocity**, and/or **high-variety** information assets that demand cost-effective, innovative forms of information processing that enable enhanced **insight**, **decision making**, and **process automation**.”
- Data also exists in different formats, like structured data, semi-structured data, and unstructured data.
- **Structured Data**: Excel Sheet
- **Semi-structured Data**: Email
- **Unstructured Data**: Pictures and Videos

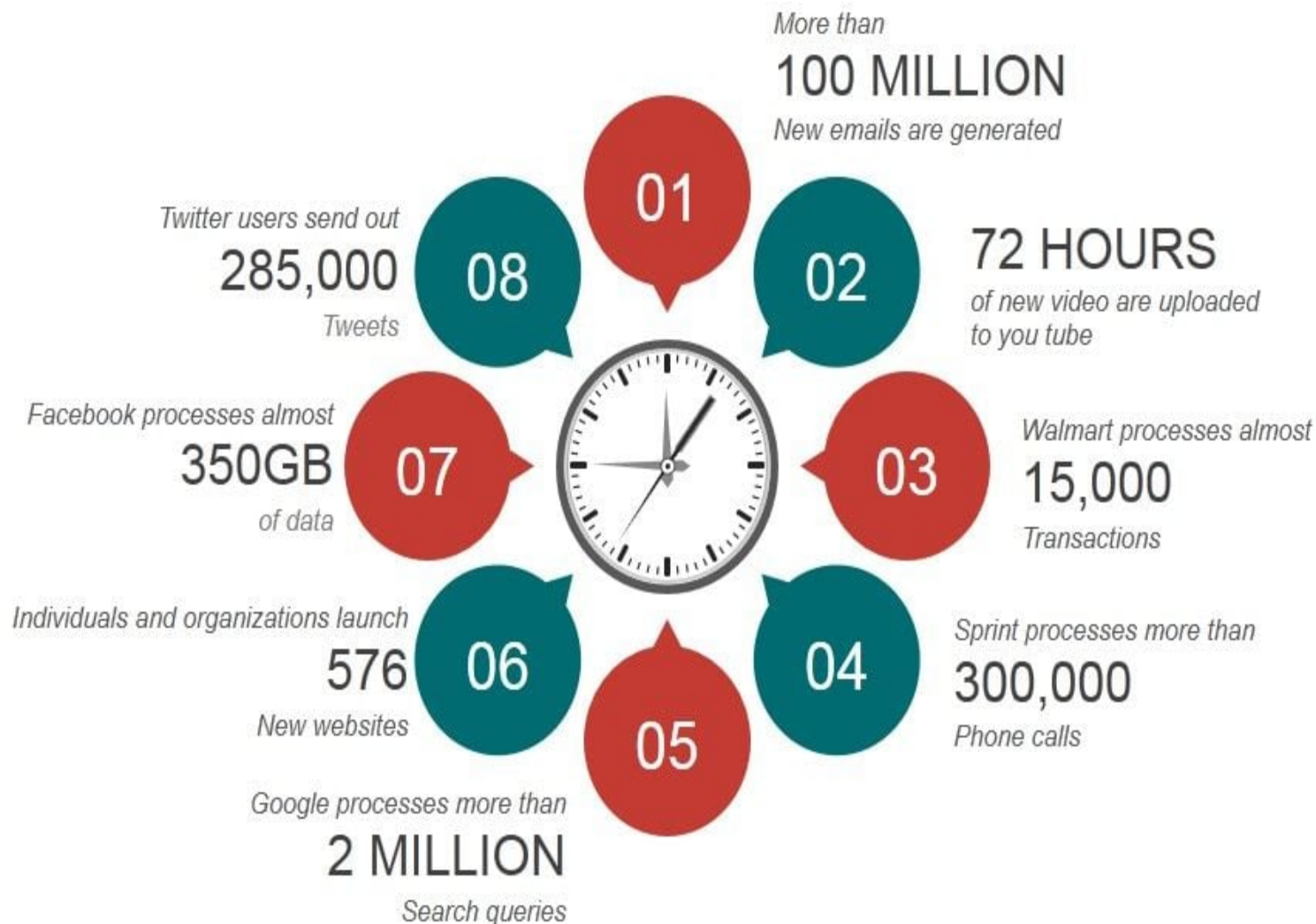


Big Data Sizes



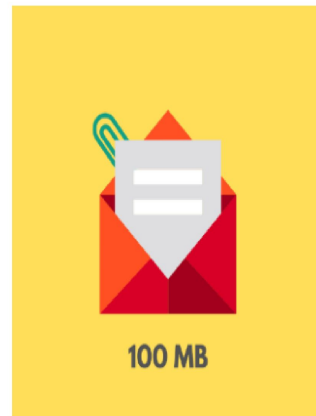


Big Data Facts-how Big Is Big Data





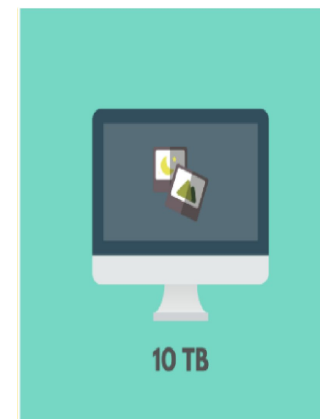
Example 1



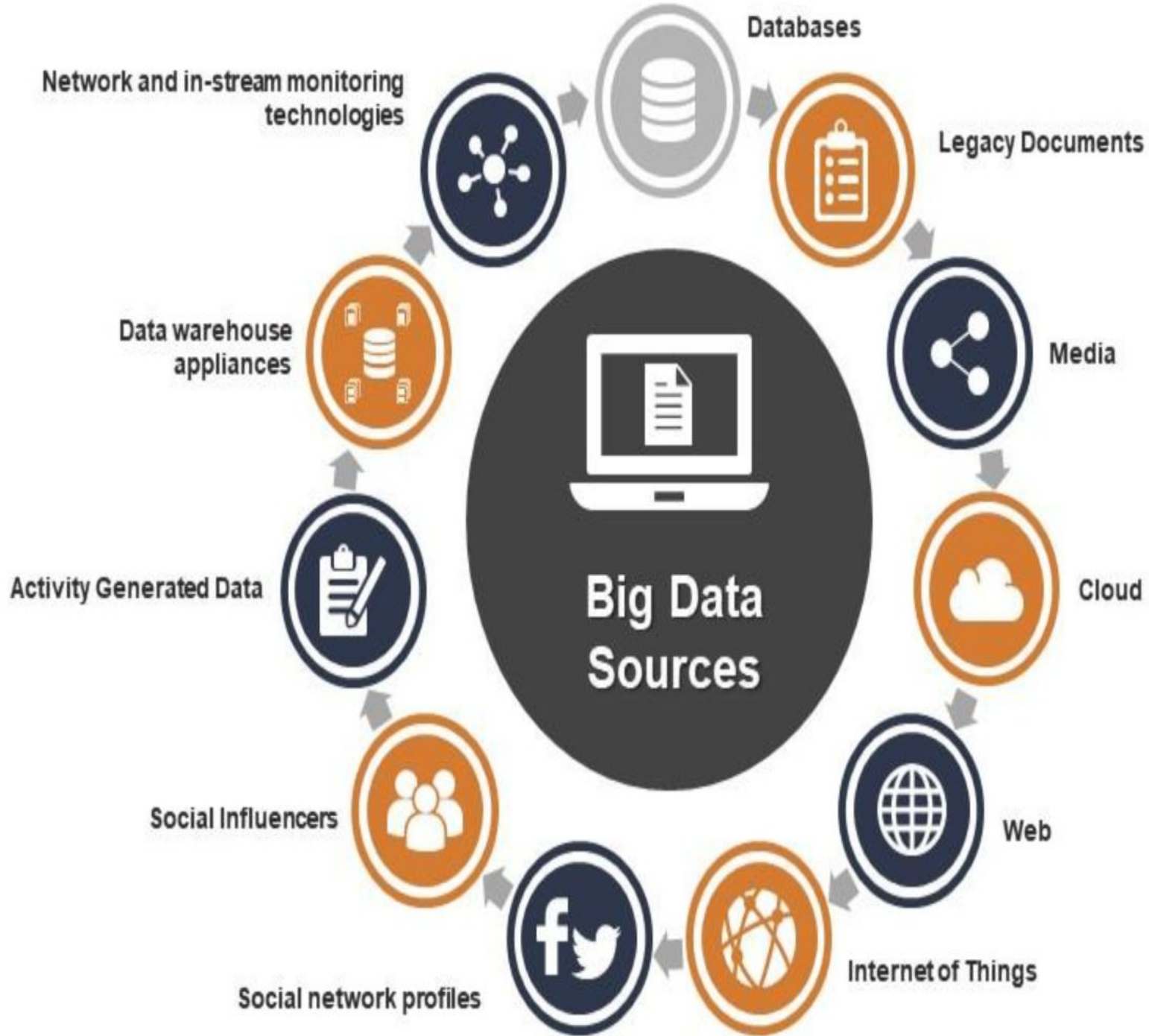
- If we try to attach a document that is of 100 megabytes in size to an email we would not be able to do so.
- As the email system would not support an attachment of this size.
- Therefore this 100 megabytes of attachment with respect to email can be referred to as Big Data.



Example 2



- Let's say we have around 10 terabytes of image files, and we want to resize and enhance these images within a given time frame.
- Therefore this 10 terabytes of image files can be referred to as Big Data with respect to processing them on a desktop computer.

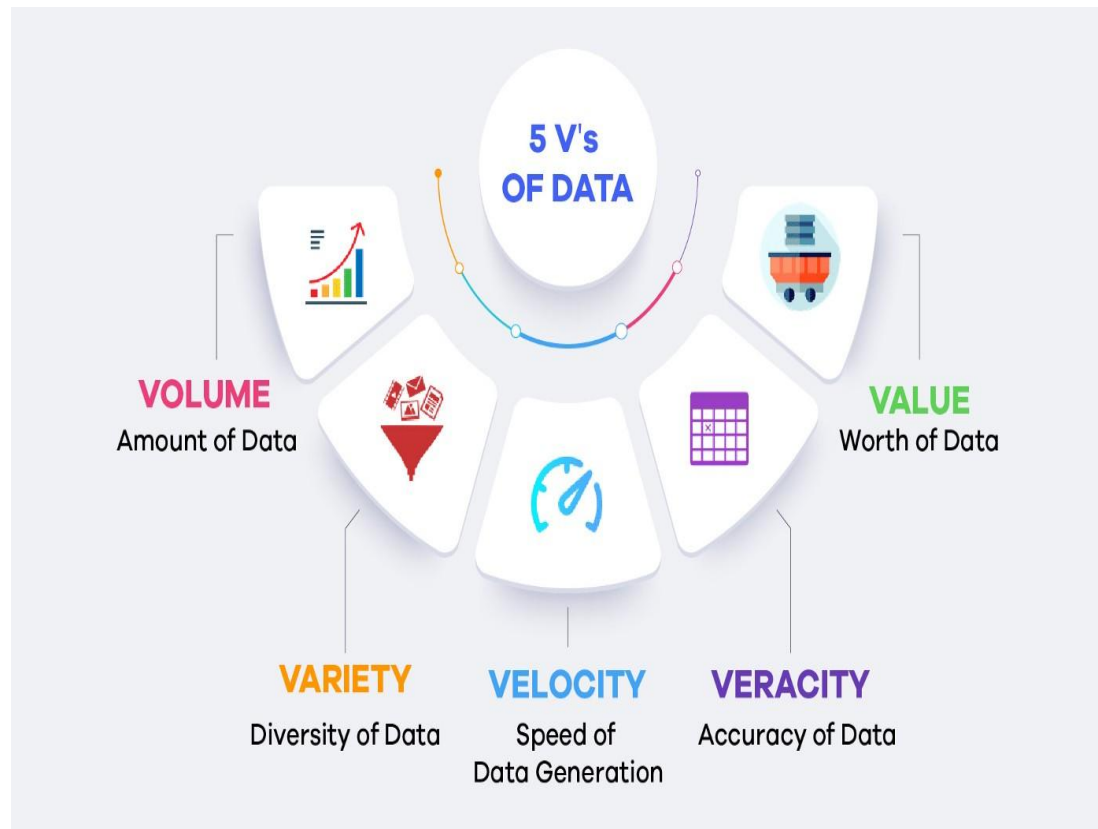


The Sources of Big Data

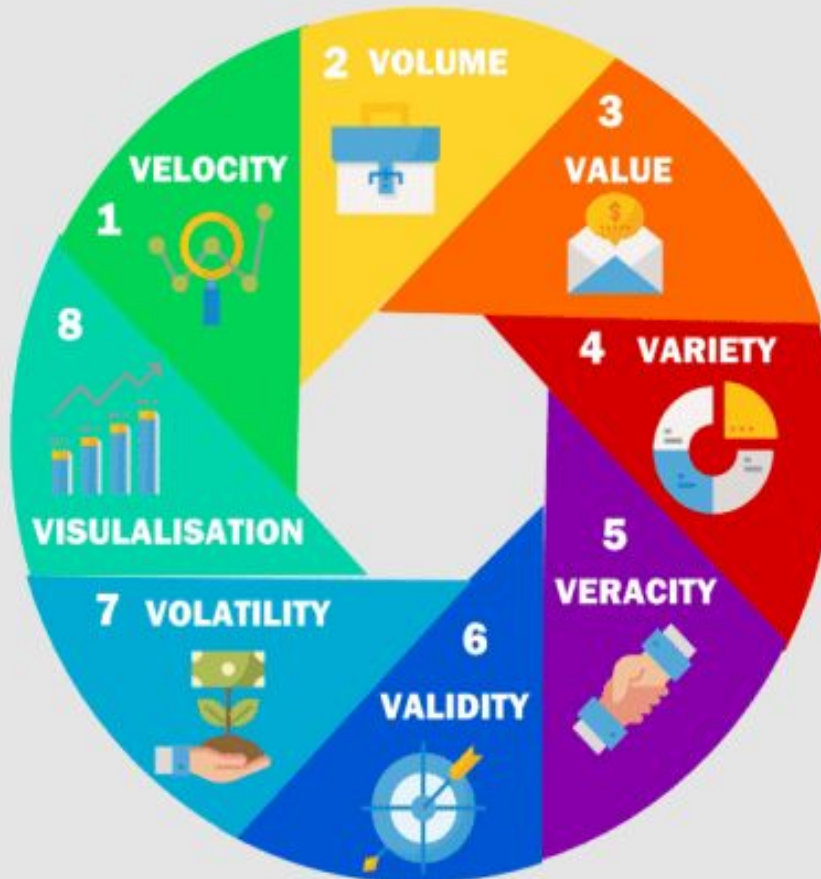
1. **Black Box Data:** This is the data generated by airplanes, including jets and helicopters. Black box data includes flight crew voices, microphone recordings, and aircraft performance information.
2. **Social Media Data:** This is data developed by such social media sites as Twitter, Facebook, Instagram, Pinterest, and Google+.
3. **Stock Exchange Data:** This is data from stock exchanges about the share selling and buying decisions made by customers.
4. **Power Grid Data:** This is data from power grids. It holds information on particular nodes, such as usage information.
5. **Transport Data:** This includes possible capacity, vehicle model, availability, and distance covered by a vehicle.
6. **Search Engine Data:** This is one of the most significant sources of big data. Search engines have vast databases where they get their data.



Characteristics of Big Data/ 5V's of Big Data



The Vs of Big Data



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1. **VELOCITY** - the speed at which data is generated, collected and analysed.

2. **VOLUME** - the amount of data generated each second. Volume is often used in reference to tools such as social media, credit cards, phones, photographs.

3. **VALUE** - this refers to the worth of the extracted data. Large amounts of data are useless unless you use it correctly.

4. **VARIETY** - this describes the different types of data generated. This term is largely used in reference to unstructured data such as images or social media posts.

5. **VERACITY** - this refers to how trustworthy data is. If the data is not accurate or of poor quality, it is of little use.

6. **VALIDITY** - like veracity this tells us how accurate the data is for its intended use.

7. **VOLATILITY** - refers to the age of the data. As fresh data is generated every hour or even minute stored data can quickly become irrelevant or historic.

8. **VISUALISATION** - describes how challenging data can be to use. Limitations such as poor scalability or functionality can impact on visualisation.

Types of Big Data

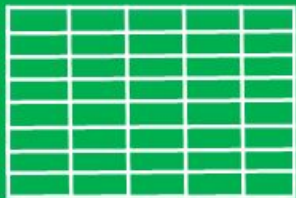
Structured

Pre-defined data models like databases

Usually text only

Easy to search and filter

Examples:
Dates, phone numbers, transaction information



Semi-Structured

Both structured & unstructured qualities

Considerably easier to analyze than unstructured data

Examples:
Emails, CSV files, JSON files



Unstructured

No Pre-defined data models

Difficult to search through

Usually stored as different types of files

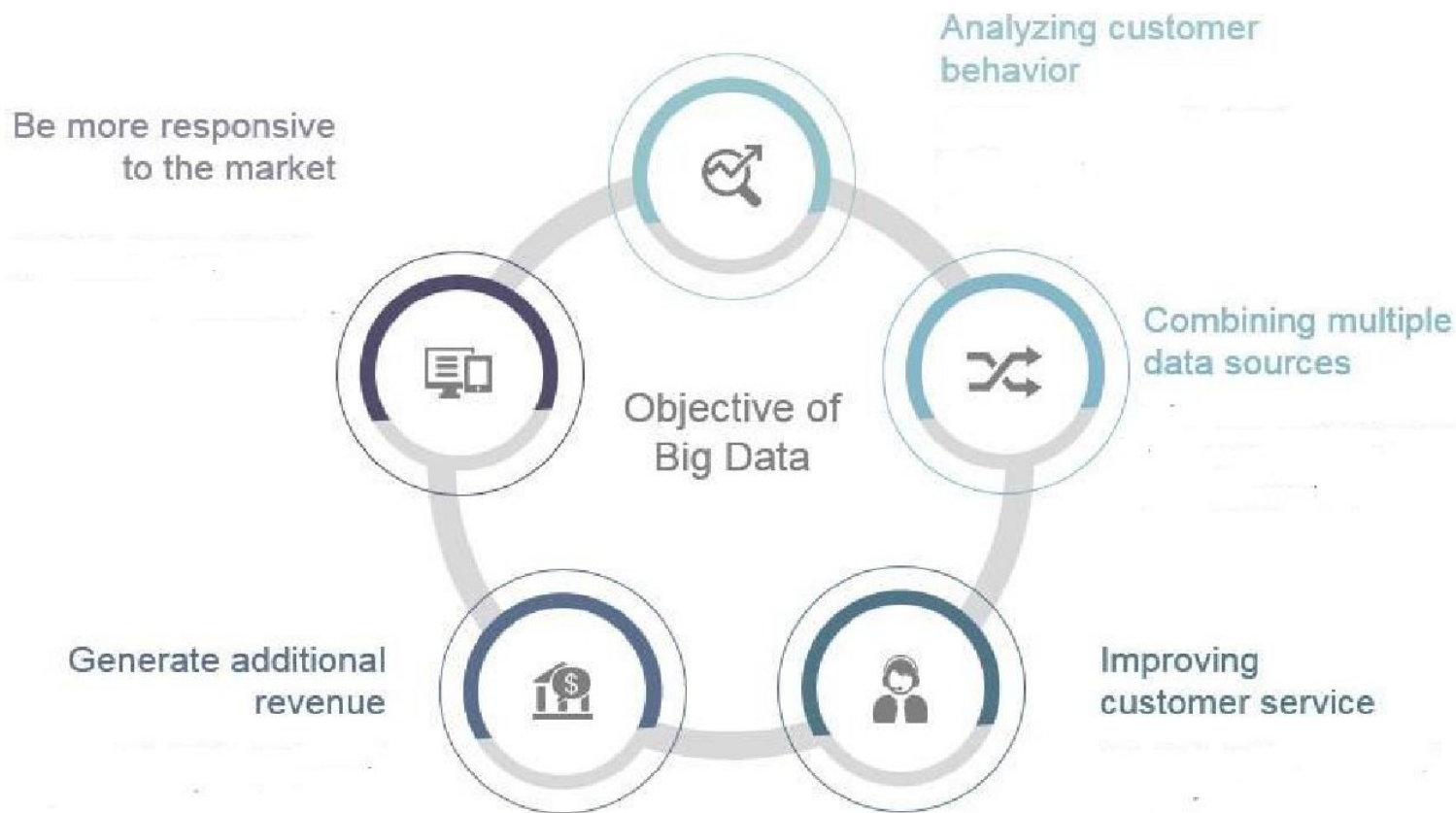
Examples:
Social media data, audio files, images





Big Data Vs Small Data

	Big Data	Small Data
Source	Data is generated from external sources i.e. Social media, device data, images, etc.	Data is generated within the enterprise i.e. from CRM system, web transactions, and financial data.
Volume	Terabytes to Zettabytes.	Gigabytes to Terabytes.
Speed	Real time.	Near real time.
Variety	Structured, Unstructured, Multistructured.	Structured, Unstructured.
Value	Complex, advanced, predictive business analysis and insights.	Business intelligence, analysis and reporting.



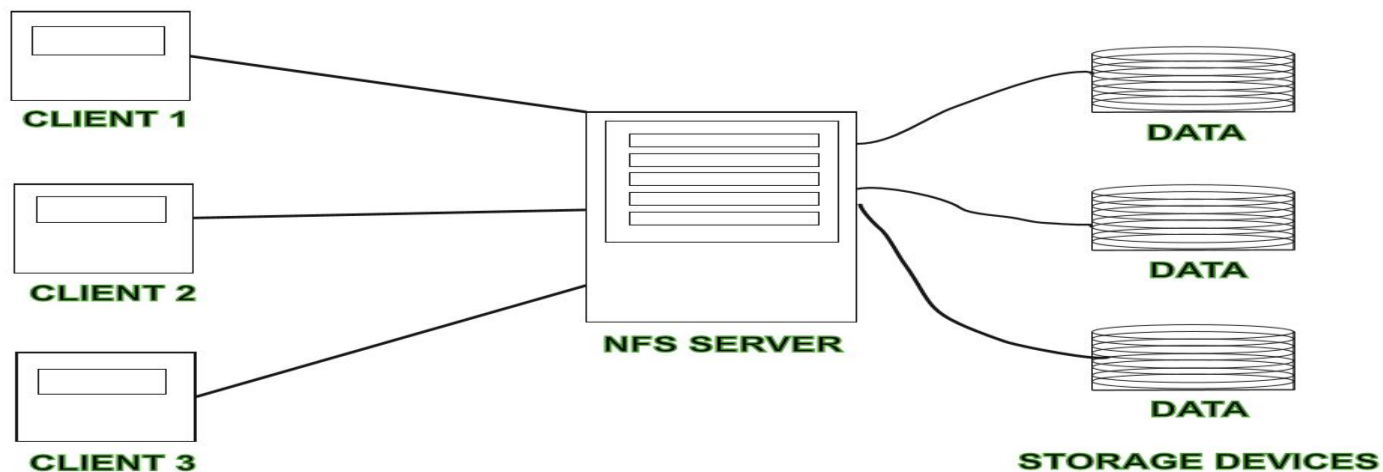
Traditional data management Approach

Traditional data systems like relational databases and data warehouses

- Clearly defined fields organized in records
- Schema-on-write
- A design to get data from the disk and load the data into memory to be processed by applications
- Structured Query Language (SQL)
- Relational and warehouse database systems often read data in 8k or 16k block sizes
- In a number of traditional siloed environments data scientists can spend 80% of their time looking for the right data and 20% of the time doing analytics

Distributed file system

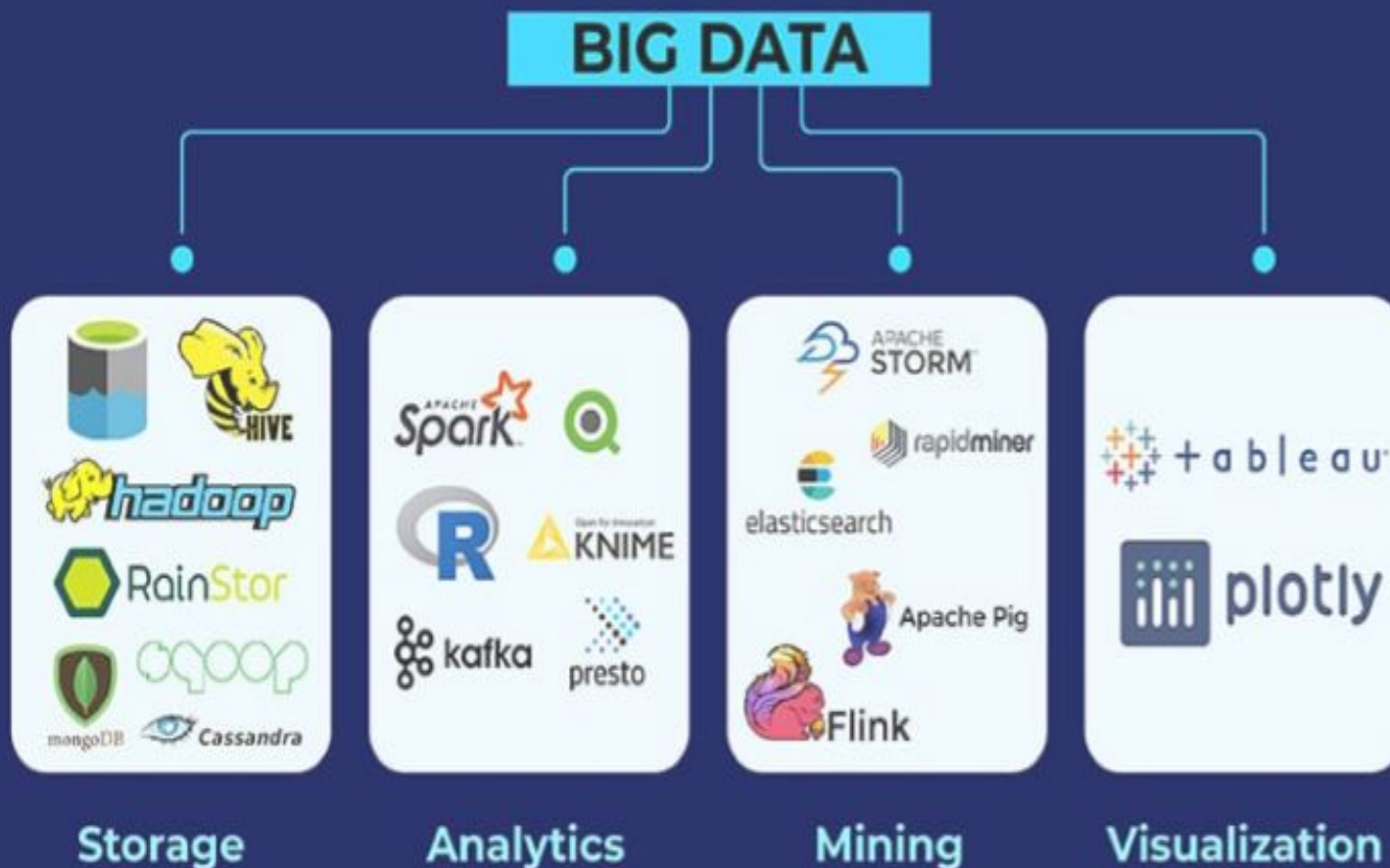
- A **Distributed File System (DFS)** as the name suggests, is a file system that is distributed on multiple file servers or multiple locations.
- It allows programs to access or store isolated files as they do with the local ones, allowing programmers to access files from any network or computer.



Big Data Approach

- **An approach that involves managing Big Data from different sources or databases.**
- **Many IT tools** are available for Big Data projects.
 - Hadoop**- Storage requirement
 - Apache Spark**- Stream Processing

Top Big Data Technologies & Techniques





Traditional I Versus Big Data Approach

Sl. No.	Traditional Data	Big Data
1.	Here the data is “Structured” data	Here the data is “Unstructured or Semi structured” data
2.	The size of the data is very small	The size is more than the traditional data size
3.	Here the data is Centralized	Here the data are distributed
4.	It is easy to work or manipulate	It is difficult to handle the data
5.	Normal system configuration is sufficient to process	High system configuration is required to process the data
6.	A traditional database tools is enough	Special kind of tools are required
7.	Normal functions are enough to manipulate the data	Requires special kind of functions to manipulate the data

INDUSTRY EXAMPLES OF BIG DATA

1. Retail and E-commerce: Role of Big Data

- **Customer Analytics:** Analyze customer data (purchase history, online behavior, social media) to understand preferences and provide personalized experiences.
- **Inventory Management:** Predict demand, minimize stockouts, and reduce excess inventory.
- **Price Optimization:** Dynamically adjust prices based on trends, competitor pricing, and demand.
Example: Personalization through Big Data enables retailers to customize product recommendations, promotions, and campaigns, driving customer satisfaction and sales.

2. Healthcare: Role of Big Data

- **Patient Care:** Monitor health, track medical records, and improve treatment outcomes with precise diagnoses.
- **Drug Discovery:** Analyze biological datasets to accelerate drug development.
- **Public Health:** Monitor disease outbreaks, track trends, and optimize resource allocation.

Example:

- **EHR & Patient Monitoring:** Analyze electronic health records (EHR) to identify patterns, enable personalized treatments, and improve outcomes. IoT devices and wearables collect real-time health data for early anomaly detection and preventive care.

3. Finance: Role of Big Data

- **Fraud Detection:** Identify suspicious transactions and patterns to prevent fraud and enhance security.
- **Risk Assessment:** Evaluate credit risks, investment opportunities, and market trends for informed decision-making.
- **Algorithmic Trading:** Leverage real-time data and machine learning for high-frequency trading decisions.

Example:

- **Fraud Detection & Risk Management:** Analyze transaction data in real-time to detect fraud. Use diverse data sources (social media, online behavior, historical data) for accurate credit risk assessment and better lending decisions.

4. Manufacturing: Role of Big Data

- **Predictive Maintenance:** Monitor equipment health in real-time to enable proactive maintenance, minimizing downtime and costs.
- **Supply Chain Optimization:** Enhance logistics and efficiency by analyzing supply chain data.
- **Quality Control:** Identify defects and improve product quality by analyzing production data.

Example:

- **IoT & Supply Chain Optimization:** IoT sensors collect real-time machine data for predictive maintenance, reducing downtime. Big Data optimizes the supply chain by analyzing raw material availability, production schedules, and customer demand for timely delivery and cost savings.



5. Transportation and Logistics: Role of Big Data

- **Route Optimization:** Optimize transportation routes to reduce delivery times and fuel consumption.
- **Fleet Management:** Monitor vehicle health, driver behavior, and safety compliance.
- **Real-time Traffic Analysis:** Analyze traffic data to manage flow and reduce congestion.

Example:

- **Predictive Maintenance & Route Optimization:** IoT sensors monitor vehicle health to predict maintenance needs, minimizing downtime. Big Data identifies efficient delivery routes, lowering fuel costs and improving delivery times.
- 4o



6. Marketing and Advertising: Role of Big Data

- **Targeted Advertising:** Deliver personalized ads to specific customer segments based on interests and behavior.
- **Social Media Analytics:** Analyze social media data to track customer sentiments, monitor brand reputation, and engage effectively.

Example:

- **Social Media Analytics & Targeted Advertising:** Track customer opinions and interactions on platforms like Twitter and Instagram to understand audiences better. Use Big Data algorithms to deliver personalized ads, enhancing campaign effectiveness and customer engagement.
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WEB ANALYTICS

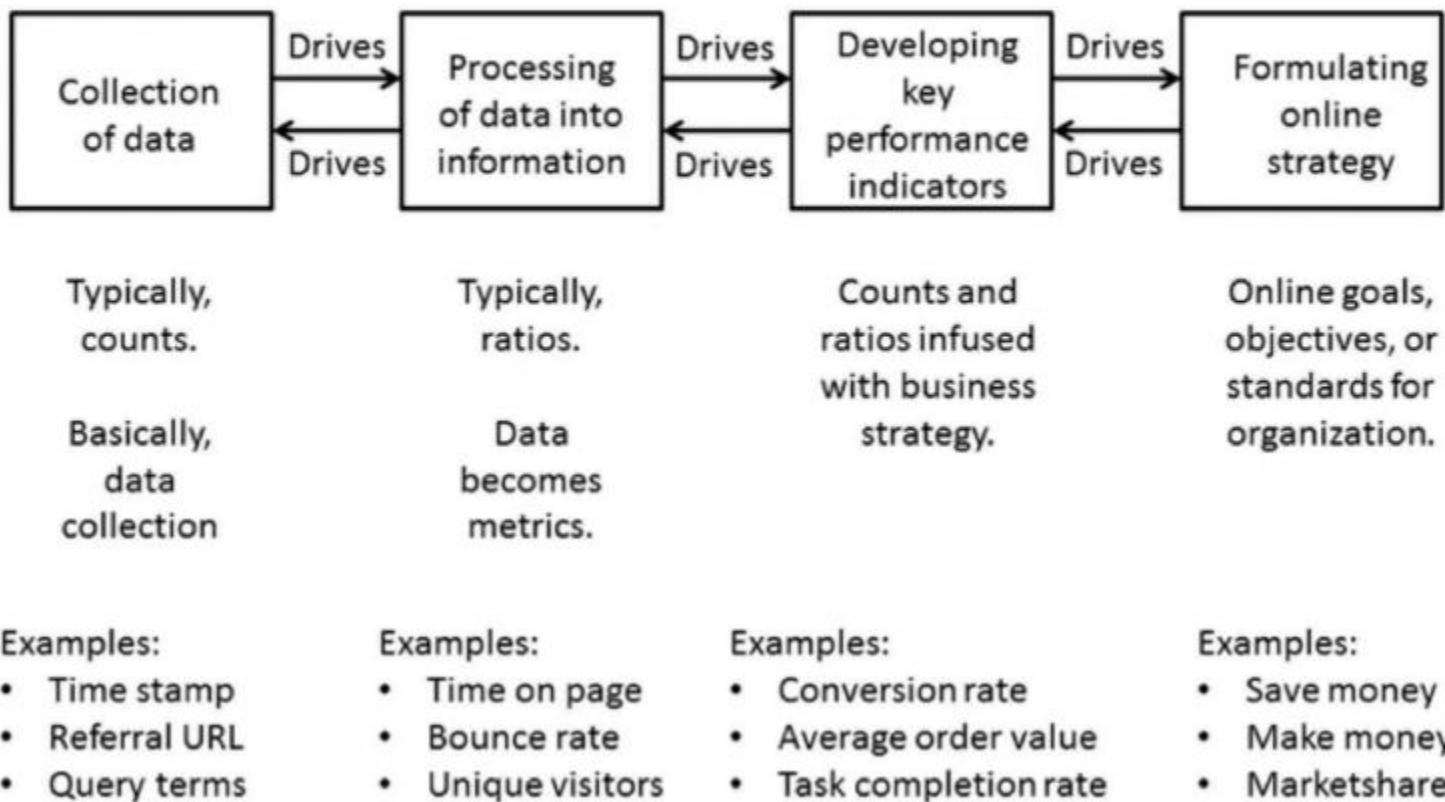
- **Web analytics** involves collecting, measuring, analyzing, and reporting data on website usage and user behavior to optimize online presence and achieve business objectives.
- **Key Components:**
- **Data Collection:** Use cookies, tracking codes, and log files to record visitor activities (e.g., page views, clicks, time spent, referral sources).
- **Data Measurement:** Quantify metrics like visitor count, bounce rate, conversion rates, and engagement levels.
- **Data Analysis:** Identify trends and insights by analyzing user behavior, navigation paths, and interactions.
- **Reporting:** Present insights through reports and dashboards, aiding in performance tracking and data-driven decision-making.



• Uses and Benefits of Web Analytics

- **Performance Optimization:** Analyze user behavior and conversion funnels to identify areas for improvement, enhancing user experience and boosting conversions.
- **Marketing Effectiveness:** Track traffic sources, evaluate marketing channels, and measure ROI to optimize marketing strategies.
- **User Experience Enhancement:** Understand user behavior and tailor content, design, and navigation to meet audience expectations.
- **Popular Web Analytics Tools**
- **Google Analytics:** Comprehensive and widely used, suitable for tracking and analyzing website data.
- **Adobe Analytics:** Advanced platform offering in-depth insights for large enterprises and e-commerce.
- **Matomo (formerly Piwik):** Open-source, self-hosted solution providing more control over data.

Basic Steps of Web Analytics Process





Application Areas of Big Data





BIG DATA TECHNOLOGIES

- Types of Big Data Technologies:

1. Operational Big Data
2. Analytical Big Data

1. Operational Big Data Technologies

- **Definition:**

Basic, day-to-day data processing technology.

- **Examples:**

- Online ticket booking (buses, trains, flights, movies)
- E-commerce platforms (Amazon, Flipkart, Walmart)
- Social media data (Facebook, Instagram, WhatsApp)
- Employee data in multinational companies



2. **Analytical Big Data Technologies:** Advanced analysis of big data for insights and decision-making.

- **Examples:**

- Stock market data
- Weather forecasting and time-series analysis
- Medical health records (monitoring patient health)
- Space mission databases

- **Categories of Big Data Technologies**

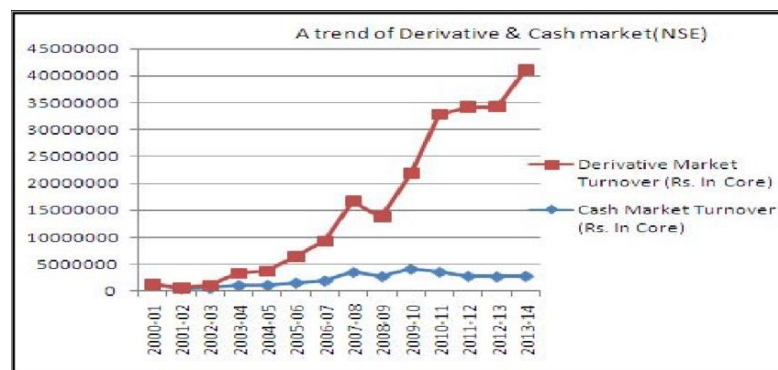
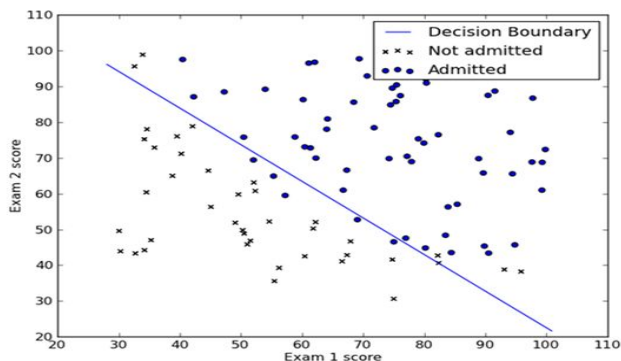
1. **Data Storage:** Efficiently storing large volumes of data.
2. **Data Mining:** Extracting valuable information and patterns.
3. **Data Analytics:** Analyzing data for actionable insights.
4. **Data Visualization:** Representing data visually for better understanding.

Big Data Technologies Overview

- **Apache Hadoop**
 - Open-source framework with **HDFS** for distributed storage and **MapReduce** for parallel data processing.
- **Apache Spark**
 - In-memory data processing for faster analytics; supports batch processing, streaming, and machine learning.
- **Apache Kafka**
 - High-throughput platform for real-time data streaming and integration.
- **NoSQL Databases**
 - Scalable databases (e.g., MongoDB, Cassandra, HBase) for unstructured and semi-structured data.
- **Apache Flink**
 - Real-time, low-latency stream processing engine for event-driven applications.
- **Apache Hive**
 - SQL-like querying framework for data warehousing on Hadoop.
- **Apache Pig**
 - High-level scripting platform for analyzing large datasets with Pig Latin.
- **Apache HBase**
 - Columnar NoSQL database for real-time read/write access.
- **Apache Storm**
 - Real-time computation system for low-latency streaming data processing.
- **Google BigQuery & Amazon Redshift**
 - Cloud-based, serverless data warehousing solutions for large-scale analytics.

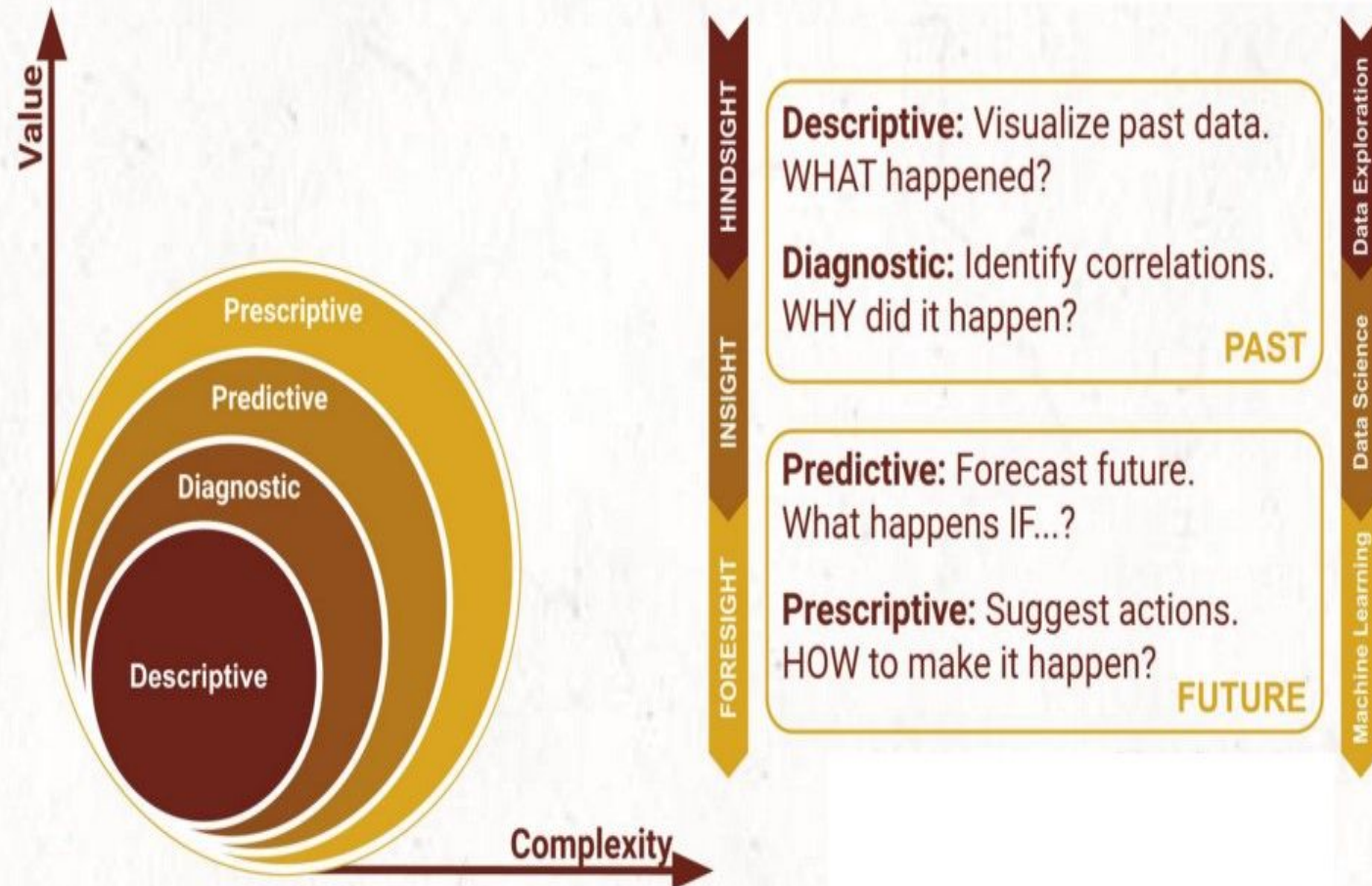
Big Data Analytics

- Big Data analytics is a process used to extract meaningful insights, such as **hidden patterns**, **unknown correlations**, **market trends**, and **customer preferences**.
- Big data analytics enables analysts, researchers, and business users to leverage big data, which was previously inaccessible and unusable, for faster and better decision-making.



Types of Big Data Analytics

4 Types of Data Analytics

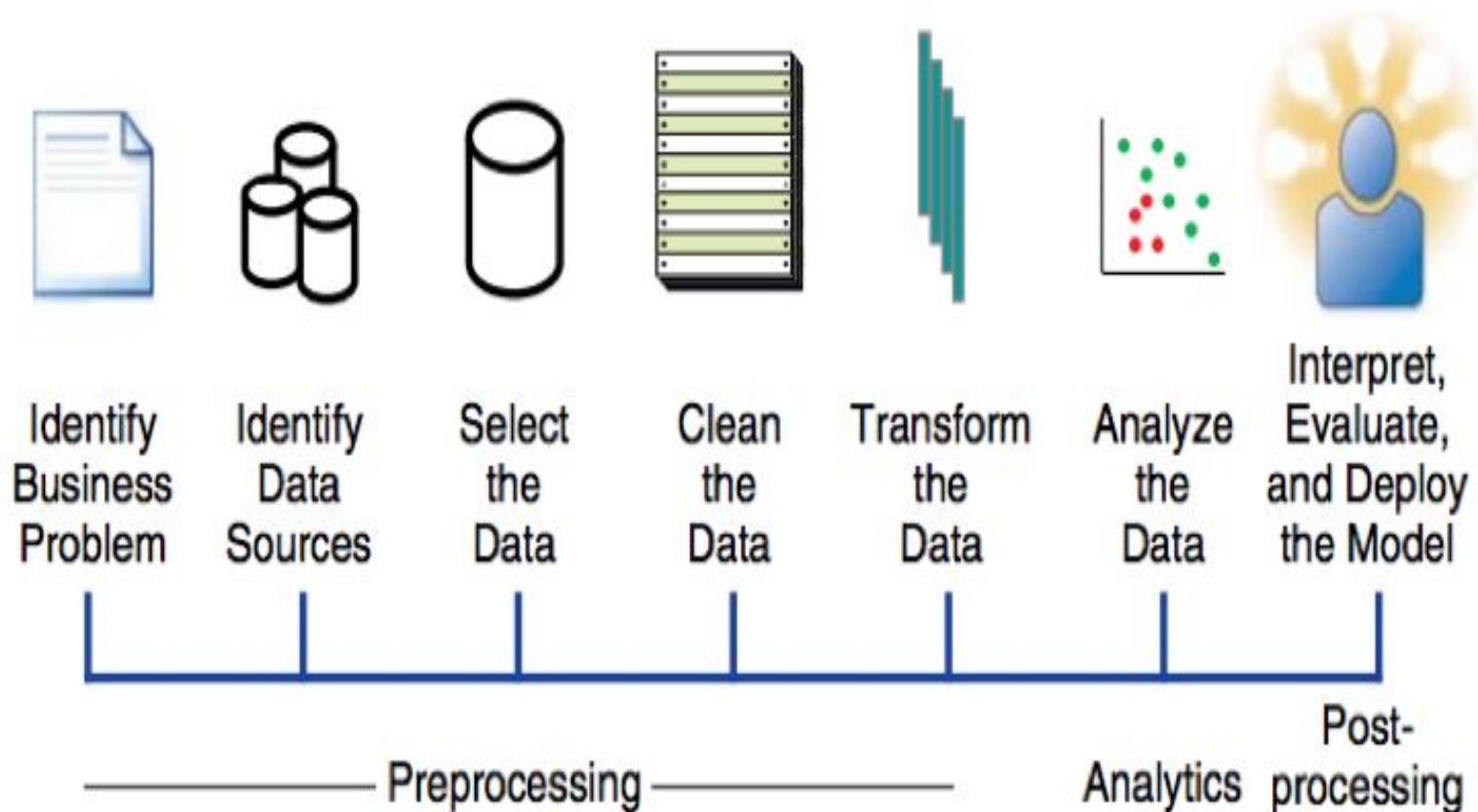


Diagnostic vs. Descriptive vs. Predictive vs. Prescriptive Analytics

The four main types of advanced analytics have some similarities, but are mainly defined by their differences. Here is a summary of how they operate:

Diagnostic	Descriptive	Predictive	Prescriptive
Uses historical data	Uses historical data	Uses historical data	Uses historical data
Identifies data anomalies	Reconfigures data into easy-to-read formats	Fills in gaps in available data	Estimates outcomes based on variables
Highlights data trends	Describes the state of your business operations	Creates data models	Offers suggestions about outcomes
Investigates underlying issues	Learns from the past	Forecasts potential future outcomes	Uses algorithms, AI and machine learning
Answers "Why" Questions	Answer "What" Questions	Answers "What Might Happen?"	Answers "If, Then" Questions

Overview of the Analytics Process Model





Benefits of Big Data Analytics

- **Data accumulation** from multiple sources, including the Internet, social media platforms, online shopping sites, company databases, external third-party sources, etc.
- **Real-time forecasting and monitoring** of business as well as the market.
- **Identify crucial points** hidden within large datasets to influence business decisions.
- Promptly **mitigate risks** by optimizing complex decisions for unforeseen events and potential threats.
- **Identify issues** in systems and business processes in real-time.
- Unlock the true potential of **data-driven marketing**.
- **Dig in customer data** to create tailor-made products, services, offers, discounts, etc.
- Facilitate **speedy delivery** of products/services that meet and exceed client expectations.
- Diversify revenue streams to boost company profits and ROI.
- **Respond to customer** requests, grievances, and queries in real-time.
- **Foster innovation** of new business strategies, products, and services.



Challenges in Big Data Analytics

- **Data Sources:** Integration challenges when it comes to combining data from sources such as social media pages, financial reports, documents by employees, customer logs, presentations, emails, etc., to create insightful reports.
- **Data Growth:** Data is growing exponentially with time, and with that, enterprises are struggling to store large amounts of data.
- **Real-time Insights:** Data sets are a treasure trove of insights. However, data sets are of no value if no real-time insights are drawn from them.
- **Data Validation:** Data validation on a Big Data scale can be rather difficult. An organization can get similar sets of data from different sources but the data from these sources may not always be similar.
- **Data Security:** Data security is usually put on the back burner, which is not a wise move at all as unprotected data can fast become a serious problem. Stolen records can cost an organization millions.
- **Big Data Skills:** Running Big Data tools requires expertise that is possessed by data scientists, data engineers, and data analysts. Although organizations are spending on recruiting professionals with such skills, organizations are also investing in training their existing staff as well.

INTRODUCTION TO HADOOP

What is Hadoop?

- Open-source distributed computing framework (developed in 2005 by Doug Cutting and Mike Cafarella).
- Enables storage and processing of large datasets in a **scalable** and **cost-effective** manner.
- Scales from single servers to thousands of machines with local computation and storage.



Key Components of Hadoop

- **Hadoop Distributed File System (HDFS):**
 - Distributed file system that splits data into blocks and replicates them across nodes for fault tolerance.
 - Highly scalable and fault-tolerant.
- **MapReduce:**
 - Programming model for parallel data processing.
 - Two stages: **Map (data transformation)** and **Reduce (aggregation)**.
- **YARN (Yet Another Resource Negotiator):**
 - Manages resources (CPU, memory) for applications.
 - Decouples resource management from job scheduling for efficiency.
- **Hadoop Common:**
 - Shared utilities, libraries, and APIs to support other components.

Why Hadoop?

- **Distributed Nature:** Handles large-scale data efficiently.
- **Fault Tolerance:** Ensures data availability despite node failures.
- **Scalability:** Grows with data volume.
- **Cost-Effective:** Uses commodity hardware for storage and processing.

Applications of Hadoop:

- **Data Warehousing:** For structured and unstructured data analytics.
- **Log Processing:** Insights from server and network logs.
- **Recommendation Systems:** E.g., e-commerce product suggestions.
- **Machine Learning:** Running algorithms on massive datasets.

Hadoop Ecosystem Tools:

- **Apache Hive:** SQL-like querying.
- **Apache Pig:** High-level scripting for data analysis.
- **Apache HBase:** NoSQL database.
- **Apache Spark:** In-memory data processing.
- **Apache Flink & Apache Storm:** Real-time data processing.

OPEN SOURCE TECHNOLOGIES

1. Data Storage Technologies

- **Hadoop HDFS:** Distributed file system for scalable storage.
- **Apache Cassandra:** NoSQL database for high availability and scalability.
- **Apache HBase:** Column-oriented database on top of Hadoop.
- **MongoDB:** NoSQL database for semi-structured and unstructured data.

2. Data Processing Frameworks

- **Apache Spark:** In-memory processing for real-time and batch processing.
- **Apache Flink:** Low-latency stream processing engine.
- **Hadoop MapReduce:** Batch processing with distributed computing.

3. Data Integration and Streaming

- **Apache Kafka:** Distributed streaming platform for real-time data.
- **Apache NiFi:** Data flow automation for data integration.

OPEN SOURCE TECHNOLOGIES

4. Data Analysis and Querying

- **Apache Hive:** SQL-like querying on Hadoop.
- **Apache Pig:** Scripting language for data transformation.
- **Presto:** Fast SQL query engine for interactive queries.

5. Data Visualization

- **Tableau Public:** Data visualization tool for insights.
- **Apache Superset:** Open-source visualization platform.
- **Grafana:** Real-time monitoring and visualization.

6. Machine Learning and Analytics

- **Apache Mahout:** Scalable machine learning algorithms.
- **TensorFlow and PyTorch:** Frameworks for deep learning on Big Data.
- **R and Apache Zeppelin:** Interactive data analytics and visualization.

7. Workflow and Resource Management

- **Apache Oozie:** Workflow scheduling for Hadoop jobs.
- **Kubernetes:** Orchestration for containerized Big Data workloads.

CLOUD AND BIG DATA

Revolutionizing Data Management

- **Key Benefits:**
 - **Data Storage & Scalability:**
 - Scalable options like Amazon S3, Google Cloud Storage, Azure Blob Storage.
 - Supports structured, semi-structured, and unstructured data.
 - **On-Demand Computing Resources:**
 - Elastic access to virtual machines and clusters for Big Data processing.
 - Handles fluctuating workloads efficiently.
 - **Data Processing & Analytics:**
 - Frameworks (Hadoop, Spark, Flink) easily deployable on cloud platforms.
 - Managed services reduce complexity.
 - **Cost Efficiency:**
 - **Pay-as-you-go** model optimizes costs for varying workloads.
 - **Real-Time Data Processing:**
 - Cloud platforms (Amazon Kinesis, Google Pub/Sub, Azure Event Hubs) enable real-time insights.

Applications of Cloud Computing + Big Data

- **Real-Time Social Media Analytics:**
 - Analyze trends and user sentiments instantly.
- **IoT Data for Smart Cities:**
 - Process sensor data for traffic, energy, and urban planning.
- **Fraud Detection in Financial Services:**
 - Real-time identification of fraudulent transactions.
- **Health Monitoring & Predictive Analytics:**
 - Cloud-based analysis for patient health and predictive care.
- **Supply Chain Management:**
 - Real-time tracking of inventory and logistics.

MOBILE BUSINESS INTELLIGENCE

Mobile BI enables access to business intelligence data and reports on mobile devices, such as smartphones and tablets, allowing real-time, data-driven decision-making anytime, anywhere.

Key Features:

- **Real-Time Data Access:** Instant access to KPIs, dashboards, and reports.
- **Interactive Dashboards:** Drill-down, filtering, and user-friendly visualizations (charts, graphs, maps).
- **Data Visualization:** Optimized for small screens for quick insights.
- **Offline Access:** Critical data available without an internet connection.
- **Secure Access:** Authentication ensures data safety.
- **Push Notifications:** Alerts for important updates.

Example: Empowering Retail Managers with Mobile BI

Scenario: A retail chain with multiple stores implements a Mobile BI solution to provide real-time insights via a mobile app.

Key Features:

- **Sales Data Integration:** Real-time access to sales data from the POS system and central repository.
- **Real-Time Sales Dashboard:** Overview of total sales, trends, and top-selling products.
- **Geospatial Analytics:** Map-based visualization of store performance by region.
- **Product Performance Analysis:** Drill-down into sales data to analyze product trends.
- **Real-Time Alerts:** Notifications for hitting sales targets or addressing underperformance.
- **Inventory Management:** Insights into stock levels and restocking needs.
- **Comparative Analysis:** Benchmarking performance across stores or regions.
- **Customer Insights:** Data on demographics, preferences, and buying patterns.

Benefits:

- **Real-Time Decision-Making:** Quick responses to sales trends and issues.
- **Improved Store Performance:** Address underperformance promptly.
- **Efficient Inventory Management:** Optimize stock levels and reduce stockouts.
- **Targeted Marketing:** Tailor campaigns based on customer insights.
- **Optimized Strategies:** Develop effective sales strategies through comparative analysis.



CROWDSOURCING ANALYTICS

Definition:

A method of gathering and analyzing data by leveraging the collective knowledge, expertise, and efforts of a diverse group of individuals or a crowd.

- **Key Aspects:**
- **Data Collection:** From sources like surveys, social media, images, and more.
- **Diverse Perspectives:** Harness insights from participants with varied expertise.
- **Problem Solving & Idea Generation:** Engage the crowd for solutions and innovative ideas.
- **Data Analysis & Processing:** Tasks like labeling data, pattern recognition, and verification.
- **Specialized Tasks:** Image/audio annotation and sentiment analysis.
- **Quality Control:** Cross-check data for accuracy and reliability.
- **Efficiency:** Scalable, cost-effective, and time-saving.

INTER AND TRANS FIREWALL ANALYTICS

1. Inter-Firewall Analytics:

Overview:

- Analyzes network traffic and security events across multiple firewalls within an organization.

Key Aspects:

- **Centralized Data Collection:** Aggregates logs and security events from all firewalls.
- **Correlation & Aggregation:** Identifies patterns across multiple firewalls to detect potential threats.
- **Threat Detection & Incident Response:** Detects cross-network security breaches and enhances response.
- **Security Policy Review:** Ensures consistent security measures across the network.
- **Performance Monitoring:** Identifies network performance issues for optimization.

2. Trans-Firewall Analytics

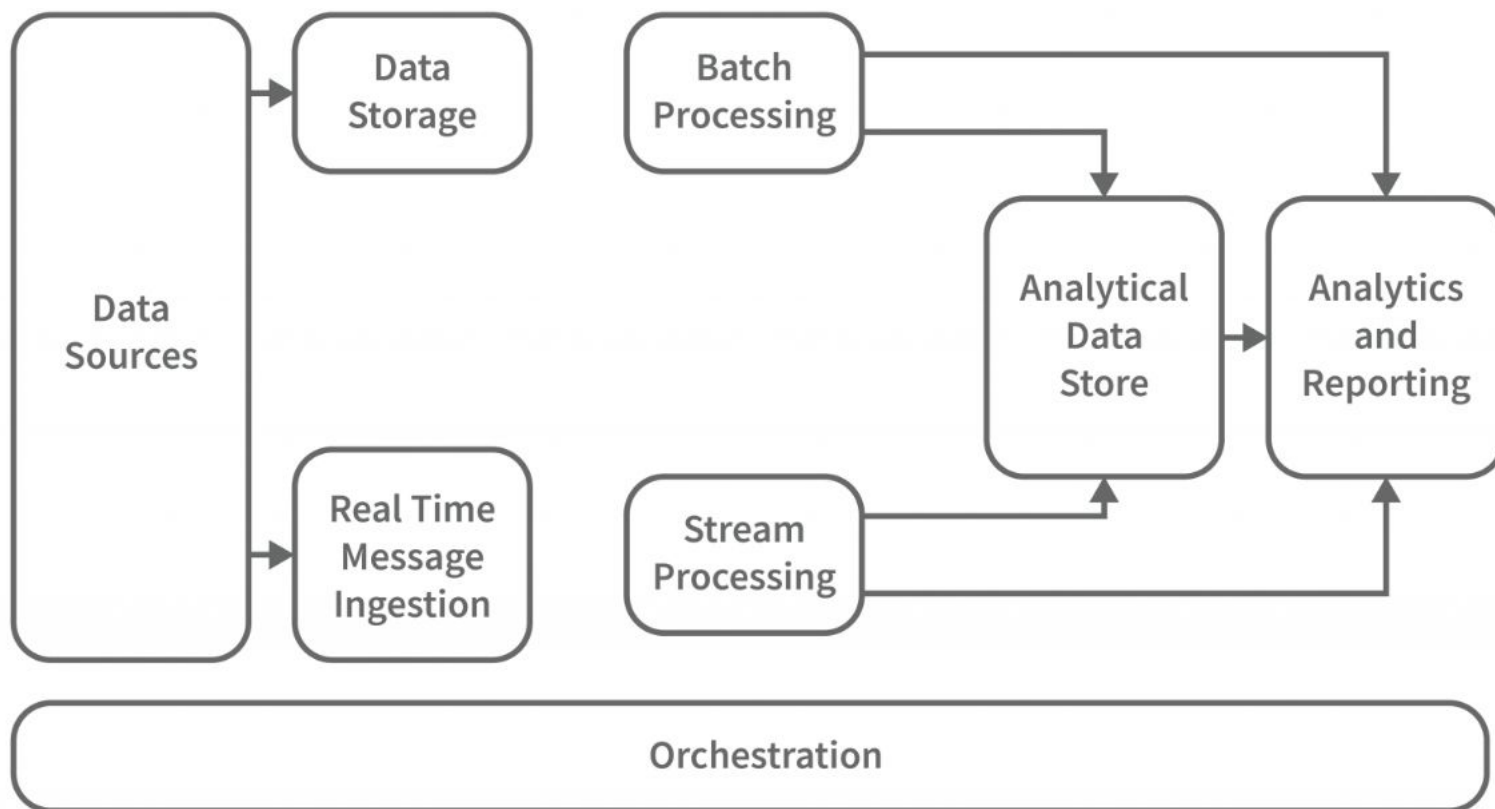
Overview:

- Focuses on deep inspection and analysis of network traffic passing through a specific firewall or set of firewalls.

Key Aspects:

- **Packet Inspection:** Examines individual packets passing through the firewall for content and payload analysis.
- **Intrusion Detection:** Detects intrusion attempts, malware, or malicious activities.
- **Policy Enforcement:** Ensures compliance with security policies and flags violations.
- **Bandwidth Management:** Identifies bandwidth-heavy applications or traffic sources for optimization.

BIG DATA ARCHITECTURE



BIG DATA ARCHITECTURE

Key Components:

- **Data Sources:** Structured, semi-structured, and unstructured data (IoT, logs, social media, etc.)
- **Data Ingestion:** Tools like Apache Kafka, Flume for batch/real-time data collection.
- **Data Storage:**
 - **HDFS** for scalable storage
 - **NoSQL Databases** (MongoDB, Cassandra) for flexible data storage
- **Data Processing:**
 - **Batch Processing:** Apache Spark, Hadoop MapReduce
 - **Stream Processing:** Apache Storm, Flink
- **Analytics & BI:** Tools like Apache Hive, Impala for querying, with BI tools like Tableau.
- **Data Visualization:** Dashboards and reporting tools for actionable insights.
- **Data Security:** Encryption, access control for compliance and privacy.

Crowdsourcing for Product Feedback: A Case Study

- **Survey Design:**
 - Structured survey on user satisfaction, features, and ratings.
- **Engaging the Crowd:**
 - Distribute the survey to diverse participants (customers, tech enthusiasts, etc.).
- **Data Collection:**
 - Collect quantitative ratings and qualitative feedback over a set period.
- **Data Analysis:**
 - Use sentiment analysis, text mining, and categorization tools to process feedback.
- **Insights & Findings:**
 - Identify strengths, weaknesses, and improvement areas of the product.
- **Comparison with Internal Data:**
 - Validate crowdsourced insights with internal customer support data.
- **Decision Making:**
 - Make data-driven decisions to enhance product features and marketing.
- **Product Iteration:**
 - Implement improvements based on feedback and launch revised version.

In every lecture, Faculty may talk on any of the following:

- **DCS & GSFCU Portal:** Remind students to use DCS, talk about it's features and instruct all to visit GSFCU website
- **Discipline, Values and Ethics.**
- **GSFC University – My University:** Emphasize the importance of taking ownership of the University.
- **Respect & Courtesy:** Remind students to be respectful to peers, teachers, and staff. Encourage polite language and behavior.
- **Punctuality:** Emphasize the importance of arriving on time to class and completing assignments by deadlines.
- **Classroom Rules:** Reinforce basic classroom rules, such as keeping mobile phones on silent and maintaining a clean environment.
- **Safety:** Remind students about safety protocols, whether it's physical safety in the classroom or digital safety when using online resources.
- **Positive Attitude:** Encourage a positive mindset, cooperation and supporting one another.

GSFCU & Values

Respect & Courtesy

Font

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Dress
Appropriately



Respect
Personal Space



Be Mindful
of Your Words



Respect
Coworkers' Time



Keep Your
Workplace Clean



Socialise with
Your Peers



Use Your
Mobile Wisely



Take
Responsibility

*The
End*